

MARCO STEFANELLI

Machine Learning Scientist | Earth Observation, Weather & Climate | Python, PyTorch & ML-Ready Data Pipelines

- Nationality: Italy
- Email: stfmrc.dev@gmail.com
- Portfolio: <https://stfmrc.github.io/MarcoStefanelli/>
- Languages: Italian (native), English (fluent), Croatian (beginner), French (willing to learn)

PROFESSIONAL SUMMARY

Machine-learning scientist with a background in physics and a PhD in Earth System and Climate Science, awarded a Marie Skłodowska-Curie COFUND postdoctoral fellowship ([SMASH](#)) to develop a neural-network-based observation operator for weather-radar data assimilation. Experienced in large-scale preprocessing of heterogeneous radar and numerical-model data; ML model design, training, evaluation and integration with a 3DVar framework; publication of reusable open-source software and TB-scale datasets. Able to lead projects from problem formulation and data preparation through experimentation, validation, implementation, and publication, while collaborating effectively across multidisciplinary scientific and operational teams.

RELEVANT EXPERTISE

- ML model development and evaluation for weather and Earth-system applications
- Climate modelling, numerical weather prediction, ocean modelling, data assimilation, and forecast validation
- TB-scale processing and curation of weather-radar, satellite, in-situ and numerical-model data
- Construction of reproducible, ML-ready datasets and preprocessing pipelines
- Management of structured and unstructured datasets
- GPU-accelerated scientific computing in Linux and HPC environments
- Open-source software, dataset documentation, and interdisciplinary collaboration

TECHNICAL SKILLS

Machine learning: PyTorch, TensorFlow, deep learning, model implementation, training and evaluation, neural networks, loss function design

Programming and data processing: Python, NumPy, xarray, GeoPandas, Rasterio, Bash, Fortran

Large-scale computing: GPU-accelerated workflows, CUDA, Linux/Unix, HPC, SLURM, PBS, LSF

Scientific data: More than 100 TB of weather-radar, satellite, in-situ, and numerical-model data, primarily in NetCDF, GRIB, and HDF5 formats

Research engineering: Git, GitHub, Jupyter, reproducible preprocessing pipelines, dataset publication and documentation, code review, dependency and environment management

PROFESSIONAL EXPERIENCE

May 2026 – July 2026

Scientific Author (Contract), Faculty of Mathematics and Physics, University of Ljubljana

- Led and co-authored a manuscript documenting the generation, processing methodology, applications, and quality characteristics of two new TB-scale datasets combining five years (2019–2023) of ALADIN model outputs with weather-radar reflectivity observations over Slovenia; the manuscript is currently under review.

April 2024 – April 2026

Postdoctoral Researcher, SMASH (Marie Skłodowska-Curie COFUND), Faculty of Mathematics and Physics, University of Ljubljana

- Led the end-to-end research workflow, from problem formulation and data preparation through model development, experimentation, validation, and publication.
- Designed and developed a neural-network-based observation operator for assimilating weather-radar observations into a 3DVar data-assimilation framework.
- Built reproducible preprocessing workflows for heterogeneous, TB-scale datasets combining weather-radar reflectivity observations with numerical weather prediction model outputs.

- Developed, trained and evaluated neural-network models in Python (PyTorch) using GPU-accelerated, CUDA-enabled workflows.
- Evaluated how alternative preprocessing and data-transformation strategies affected neural-network accuracy and subsequent data-assimilation performance.
- Collaborated with scientists from complementary disciplines at SISSA to address methodological and technical challenges in implementing the neural-network observation operator.
- Worked with researchers at the Slovenian Environment Agency to integrate the resulting observation operator into a 3DVar data-assimilation framework.
- Released open-source code and datasets to support reproducibility and further scientific use.
- Led and co-authored the preprint “A Neural Network-Based Observation Operator for Weather Radar Data Assimilation” (manuscript currently under review).

February 2023 – March 2024

Researcher, Euro-Mediterranean Centre on Climate Change (CMCC), Italy

- Developed a novel recursive filter algorithm for 3DVar and evaluated data-assimilation methodologies for high-resolution unstructured-grid ocean models.
- Processed and analysed large (TB-scale), heterogeneous observational datasets, including Argo profiling floats and satellite observations.
- Designed validation workflows comparing numerical-model outputs against observational data.
- Built reproducible Python and HPC workflows for large scientific datasets and operational modelling systems.
- Collaborated with the Data Science Group at Imperial College London on methodological aspects of neural-network development and evaluation.
- Led and co-authored the publication “Data Assimilation for Advanced Cross-Scale Unstructured-Grid Ocean Modelling.”

April 2018 – April 2019

Researcher, The Abdus Salam International Centre for Theoretical Physics (ICTP), Italy

- Designed computational experiments to study the effect of urbanisation on the West African Monsoon using a regional climate model.
- Processed and analysed large (TB-scale) regional climate-model and observational datasets, including precipitation and satellite measurements.
- Built Python/Linux workflows to compare, validate, and interpret model outputs against observations.

EDUCATION

• *November 2019 – January 2023*

PhD, Future Earth, Climate Change and Societal Challenges — University of Bologna & CMCC, Italy

Dissertation: [Data assimilation for advanced cross-scale unstructured-grid ocean modelling](#).

• *September 2014 – March 2018*

MSc, Earth System Physics — University of Trieste, Italy

Dissertation: A study of urbanisation effects on the West African monsoon.

SELECTED PUBLICATIONS

- Stefanelli, M., Jansen, E., Aydogdu, A., Federico, I., Coppini, G. (2025). [Data Assimilation for Advanced Cross-Scale Unstructured-Grid Ocean Modelling](#). Frontiers in Marine Science.
- Stefanelli, M., Zaplotnik, Z., Skok, G. (2026) Preprint (arXiv): [A Neural Network-Based Observation Operator for Weather Radar Data Assimilation](#) (under review).

OPEN-SOURCE SOFTWARE

- [Radar-ALADIN data preprocessing](#) - Python/Bash
Reusable data preprocessing workflow for combining radar observations and numerical-model data for ML use.
- [3DVar neural network-based observation operator](#) - Python
Implementation of the NN observation operator developed during the Marie Skłodowska-Curie (SMASH) project.

- [3DVar integration of a neural network-based observation operator](#) - Python
Implementation connecting the learned observation operator with a data-assimilation framework.

PUBLISHED DATASETS

- [LISCA-ALADIN HNN](#)

Curated ML-ready dataset (NetCDF) for neural-network development and evaluation covering 2019–2023.

- [2019](#) | [2020](#) | [2021](#) | [2022](#) | [2023](#) - **ALADIN-LISCA RADAR LOWEST ELEVATION ANGLE**

Five annual datasets containing ALADIN model outputs (GRIB) and lowest-elevation-angle radar reflectivity (HDF5) observations.

SELECTED PRESENTATIONS

- European Geosciences Union (EGU), Vienna (2022 and 2025)
2022: <https://meetingorganizer.copernicus.org/EGU22/EGU22-4741.html>
2025: <https://meetingorganizer.copernicus.org/EGU25/EGU25-420.html>
- European Meteorological Society (EMS) Annual Meeting, Ljubljana (2025)
Video: <https://meetingorganizer.copernicus.org/EMS2025/EMS2025-28.html>
- Slovenian Meteorology Society, Ljubljana (2025)
Video: <https://www.youtube.com/watch?v=kIYwkgOSsGA>

ADDITIONAL INFORMATION

Science communication: Developed and delivered educational activities on climate change, scientific literacy, and misinformation for secondary-school students. Created an interactive [JavaScript tool](#) based on the Lorenz-63 model to demonstrate deterministic chaos and how data assimilation reduces forecast uncertainty.